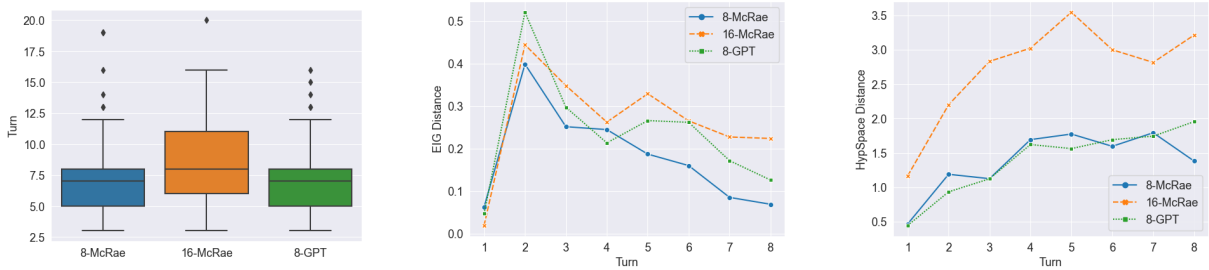

Figure 3: Increasing the candidates (**Blue vs. Orange**) causes the generation of longer dialogues (left) – maintaining a similar distance to the optimal model's AQ (8.67 vs 4.5); does not impact much the questions' EIG (middle), while it makes the guessing of the candidate by turn harder (right). Using candidates space structure based on GPT-norms (**Blue vs. Green**) does not impact any measure.

dialogue history; this weakness could impact its ability to stop asking questions when it has reached the singleton set.

Search efficiency By construction the optimal agent asks on average 3.5 questions per game, whereas the baseline asks on average 4 questions. ChatGPT-Q asks way more questions per game (7.24) compared to both models (Table 1). Such difference is statistically significant based on a Wilcoxon signed-rank test (e.g., wrt the optimal agent, $p < 0.001$). Moreover, ChatGPT-Q asks unnecessary questions (UQ) (questions asked after the singleton set has been reached) in 56.67% of games – 29.29% of its questions are unnecessary (See Figure 2, right for the distribution of UQ.) Summing up, in terms of search efficiency, ChatGPT's behavior is similar to that pre-scholar children who tend to not stop asking questions once there is only one item left in the hypothesis space.

6.2 Changing the games

Figure 3 illustrates how ChatGPT-Q's performance is effected by the Hypothesis space size (Blue vs. Orange) and of the features used to build it (Blue vs. Green). Hence, it compares the results discussed above (McRae-8) with those obtained in the other two settings: McRae-16 and GPT-8. In particular, it shows the comparison based on the number of questions per game (left), the distance between ChatGPT-Q and the optimal agent in terms of EIG (middle), and the average symmetric difference between the ground-truth hypothesis spaces update, based on ChatGPT-Oracle, and the one generated by ChatGPT-Guesser.

ChatGPT-Q on 16-McRae Given the difficulty the model has in keeping track of the space update, we expect that by increasing the number of candidates ChatGPT-Q' s performance will decrease. The results are not clear-cut: by moving from 8 to 16 candidates, the optimal agent would have an increase of 1 question per game, while ChatGPT increases of 1.43; the difference in terms of EIG is low, while ChatGPT-Guesser's performance deteriorates.[8]

ChatGPT-Q on 8-GPT The games built out of GPT-norms should reflect the model knowledge representation, therefore we expect that on the 8-GPT games ChatGPT-Q performance will increase. Instead, for none of the measures the difference is significant. This suggests that the feature norms used to build the hypothesis spaces do not impact the model's performance.

6.3 Changing the prompt

To further understand what causes ChatGPT inefficient strategy, we would need to run an ablation study by isolating the various processes that should be beyond the question generation. To simulate such study, we compare the set of results discussed so far on McRae-8 obtained by ChatGPT-Q simply prompted to play the game, with those obtained by ChatGPT- Q-stepwise, the model that is explicitly told to update the space turn by turn. Our results show that ChatGPT-Q-stepwise gets closer to the optimal model: it asks fewer questions per game (Figure 4, left) compared to ChatGPT-Q (6.4 vs. 7.24), the questions' EIG is higher across all the turns (Figure 4, middle), and it is more precise when updating the hypothesis space (Figure 4, right). This finding confirms the conjecture that ChatGPT main weakness lies in its difficulty in *mentally* updating the hypothesis space.

[8] Games generated on 16-McRae are significantly longer than those generated on 8-McRae, based on a Mann-Whitney U test ($p < 0.001$).

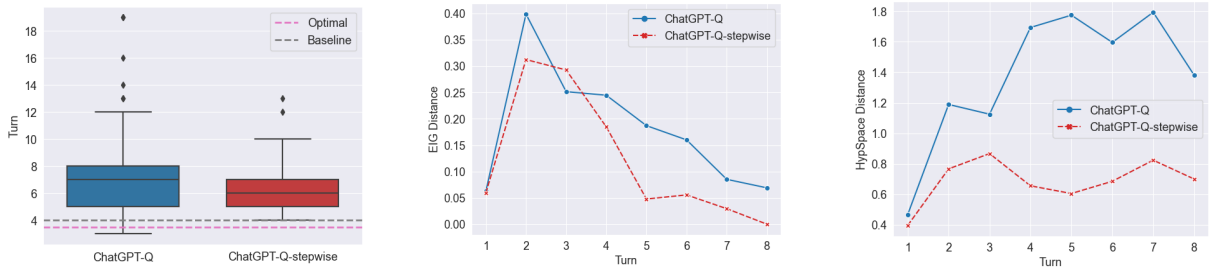

Figure 4: Changing the prompt improves the Questioner performance on all the measures, bringing it closer to the optimal model in terms of number of turns and EIG, an to the ground truth hypothesis space stepwise reduction.

ChatGPT-Q			
	CQ	TQ	SG
McRae-8	13.50	25.15	12.22
McRae-16	14.74	18.59	13.33
GPT-8	13.22	26.90	8.89
ChatGPT-Q-stepwise			
	CQ	TQ	SG
8-McRae	8.42	26.78	7.78

Table 2: Quality of the dialogue: Contradictory Questions (CQ) decrease when ChatGPT is asked to update the space explicitly. Yet, it is still unaware of Spoiled Games (SG).

7 Qualitative Analysis

In this section, we further dive into the quality of the dialogues generated by ChatGPT. First of all, we inspect whether it asks questions that do not reduce the space at all (trivial questions, TQ) or refer to candidates that have already been excluded in previous turns (contradictory questions, CQ). As we can see from the statistics reported in Table 2, ChatGPT is rather coherent through the dialogues; yet, the number of trivial questions is higher than what one would expect from a rational agent. In all the different settings, we observe a low peak in terms of EIG at the second turn. To understand the reason, we look into the 8-candidates sets (McRae and ChatGPT): 39.44% of the second questions are uninformative (EIG=0), with a large majority of trivial questions (92% of the uninformative). Interestingly, neither the size of the space nor the norms used to build it impact the number of trivial and contradictory questions. Instead, the coherence of the dialogues improves when the prompt is changed and ChatGPT is asked to update the space stepwise before asking the next question (ChatGPT-Q-stepwise on 8-McRae).

By inspecting the dialogues, we realized that in all the settings (8 vs 16 cds, McRae vs. ChatGPT-norms, explicit vs. implicit-update), there are games in which ChatGPT continues asking questions even when the Answerer has accidentally revealed the target (Spoiled games, SG – see an example in the Supplementary Material). This suggests that the model is pretending to play the game without having actually grasped the actual purpose of it. Most probably, a spoiler would not pass unobserved by a 4Y-child.

8 Conclusion

Our work shows that ChatGPT is able to identify superordinate features shared by items and ask questions that efficiently reduce the hypothesis space. At the first turn, it is close to an optimal agent using a half-split search. In later turns, however, it has difficulties making questions with respect to the updated space of the hypothesis. This weakness might be behind the high number of games in which it keeps on asking questions even though the dialogue history had led to identifying a possible target. We conjecture this behavior is not due to the lack of knowledge required by the game since it is displayed not only within the games based on McRae norms but also on those built out of GPT feature norms. Our conjecture is reinforced by the increased performance reached by the model when prompted to explicitly update the space before asking the next question. In this setting, the dialogue becomes shorter with fewer contradictory questions. Yet, even in such scenario, it does not notice when the Answerer reveals the target accidentally. Our results call for attention to modeling the human ability to keep a *mental scoreboard*, echoing what stated in Lewis (1979); Madureira and Schlangen (2022); Mazuecos et al. (2021). Finally, our work relates to the Chain-of-Thought (Wei et al., 2022) and similar prompting strategies, which we plan to investigate in the future.

9 Limitation

The backbone of the hierarchical space we built are feature norms. For the first level split, we used superordinates which by definition are disjoint. For the second level, we used all other feature norms by making sure that the feature that is shared by a subset is not listed in any of the members of the other subset; this process does not guarantee disjointedness of the two subsets. McRae features norms associated to a concept should be salient for it, while the absence of a feature from the list could be either because the feature does not hold for that concept or because it is not salient. Nevertheless we choose to use McRae-norms because they reflect human representation of the world and gave us the possibility of having a straight comparison with the games built out of ChatGPT-norms – comparison which shows that the knowledge used for building the hypothesis spaces does not impact the model's performance. We evaluated the model also on games built with taxonomic relations extracted from WordNet for both levels of the hierarchy: the patterns are very similar to those obtained with McRae- and GPT-norms (See the Supplementary Material for details.)

A second limitation is due to ChatGPT being a closed-source model, for which the exact training data is not known. We leave for future work the study of a LLM open source. Finally, we have not compared the model results on those that humans playing the games would achieve, instead we rely on the results obtained within the Cognitive Science literature about the 20-Q game.

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